NICOLETTE MILLER

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CAREER SUMMARY

Full stack engineer with 6+ years building production clinical software inside UCSF's health system, promoted twice. I own 5 production dashboards on an Epic-integrated (SMART on FHIR) platform used by 30+ clinicians across 8 departments. I partner with clinicians from discovery through deployment and adoption.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, Ruby

Frameworks & Libraries: Flask, Ruby on Rails, D3.js, jQuery

Tools & Platforms: Docker, Git, Postman, Unix/Linux, PostgreSQL, MySQL

Practices: EHR Integration (SMART on FHIR), HIPAA/PHI, RESTful APIs, Data Visualization, UX Research

PROFESSIONAL EXPERIENCE

Application Engineer III · UCSF Weill Institute for Neurosciences, Bove Lab · July 2023 – Present

SF, CA · Promoted Application Engineer II → III (July 2023)

- Own 5 of 9 production dashboards (plus 2 in development) on BRIDGE, a clinician-facing platform integrated with Epic via SMART on FHIR and used by 30+ clinicians across 8 departments. Current work includes perimenopause, adolescent depression, TBI, and inpatient TBI dashboards.
- Sole engineer responsible for BRIDGE for 9 months after my mentor's departure, keeping it reliable in production while continuing feature development.
- Integrated 7 of BRIDGE's real-time data sources in Python/Flask, including Epic EHR data, clinical prediction models, REDCap and Seizure Tracker, pulling and normalizing data at launch with no persistent storage of PHI. Currently integrating additional in-house prediction models.
- Drove hiring of the team's current tech lead: co-wrote the job description, screened candidates, interviewed, and made the hiring recommendation.
- Partner with clinicians on discovery and workflow scoping for new dashboards, currently scoping the expansion of the orthopedic dashboard into a spine dashboard.
- Developed interactive D3.js data visualizations — complex charts, graphs, and longitudinal data tables — enabling clinicians to interpret patient trends at a glance.
- Maintain Open MS Bioscreen, a patient-facing application for self-reported health metrics with 600+ registered accounts, overseeing feature development and production reliability.
- Co-author on multiple peer-reviewed publications; first author on one, with a second first-author paper in progress.

Application Programmer · UCSF Weill Institute for Neurosciences, Bove Lab · Jun 2020 – Nov 2021

- Developed dynamic, data-driven surveys in Ruby on Rails for clinical research data collection.
- Refactored front-end interfaces to improve accessibility for patients with neurological conditions, incorporating user feedback and accessibility best practices.

EDUCATION

B.S. in Computer Science · University of California, Santa Cruz · Santa Cruz, CA